| Inch/Pound |

MIL-I-42073
25 May 1990

MILITARY SPECIFICATION

INSERT, OPTICAL, CHEMICAL-BIOLOGICAL MASK, XM-40

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers one type and class of optical inserts for use with Mask, Chemical-Biological, XM-40, by personnel requiring visual corrections.

2. APPLICABLE DOCUMENTS.

2.1 Government Documents.

2.1.1 Specifications, Standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 6540

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SPECIFICATIONS

Federal

QQ-C-586	-	Copper-Nickel-Zinc Alloy; Rod, Shapes and Flat Products with Finished Edges (Flat Wire, Strip, and Bar).
QQ-W-321	-	Wire, Copper Alloy.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.

Military

MIL-L-10547	-	Liner, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	-	Metals: Test Method.

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads.
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging of.

(Unless otherwise indicated copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWING

Defense Personnel Support Center (DPSC)

24554 Insert, Optical, XM-40, Mask

Copies of these drawings may be obtained from the Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.4), the contractor shall furnish a sample for first article inspection and approval (see 4.3.1 and 6.4).

3.2 Components. The optical insert shall consist of two frames; a binocular lens subassembly and two temporal strut housings, each with temporal strut screw to support the binocular lens subassembly. The binocular lens subassembly shall consist of two eyewires, each with eyewire clamp and eyewire screw, two temporal struts, two bridge blocks, and a bridge.

3.3 Materials. When materials to be used are not specifically designated, those used shall be of the highest commercial quality and entirely suitable for the purpose intended.

3.3.1 Nickel silver parts. The eyewires, temporal struts, bridge blocks, temporal strut housing, frames and eyewire clamps shall be nickel silver conforming to copper alloy No. 745 or No. 752 of QQ-C-586 or QQ-W-321. As an alternate, the bridge blocks shall conform to copper alloy No. 752 and the temporal strut housings to alloy No. 792 of QQ-W-321. The eyewire screw shall conform to copper alloy No. 752 of the same specifications. As an alternate, eyewire screw shall be made of No. 303 stainless steel. The temper shall be suitable for the purpose intended.

3.3.2 Corrosion-resisting steel parts. The bridge and the temporal strut screw shall be of corrosion-resisting steel. The bridge shall be type 302 and the temporal strut screw shall be type 302 or 303. Wire for the bridge shall be spring tempered.

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3.3.3 Plastic parts. The covering of the insertion (or removal) tabs shall be black polyvinyl chloride plastic. It shall be 100 percent virgin material. Re-processed material shall not be used.

3.4 Style, design, and construction. The optical inserts shall conform to style, design, and construction as shown on DPSC Drawing 24554. Inserts shall be compatible with Mask, Chemical-Biological, XM-40. All soldering, except soldering of the bridge spring to bridge block (part no. 3), shall be accomplished with hard solder. Soldering of bridge spring to part no. 3 shall be accomplished with hard or soft solder.

3.4.1 Frames. The frames shall be made of 0.080 inch diameter drawn wire, as shown on the drawing, to fit the interior perimeters of the eye lenses of the mask. The vinyl covering on the insertion (or removal) tabs shall be as shown on the drawing and shall be so applied as to firmly adhere to the tabs.

3.4.2 Eyewires. The eyewires shall be made of rolled eyewire stock, 0.082 inch wide by 0.055 inch thick, and shall be grooved to adequately hold beveled ophthalmic lenses as shown on the drawing. If requested, a metal flat master lens template will be available on loan from the procuring activity. This template is to be used for lens fit testing in paragraph 4.3.3.

3.4.3 Temporal struts. The temporal struts shall be made of 0.078 inch diameter drawn wire and shall be attached to the eyewires as shown on the drawing.

3.4.4 Temporal strut housings. The temporal strut housings shall be attached to the frames in one position as shown on the drawing.

3.4.5 Bridge. The bridge shall be a close-coiled helical spring, 0.143 inch outside diameter, 0.027 inch diameter wire, and shall be attached to the bridge blocks as shown on the drawing.

3.4.6 Tolerances. Tolerances on all dimensions shall be as shown on the drawing, unless otherwise specified.

3.5 Unit. A unit shall consist of a complete assembly of two frames, each with a temporal strut housing, and a binocular lens assembly, as shown on the drawing. The lenses for the eyewires shall not be furnished.

3.6 Finish. All metal parts of the complete insert shall be nickel-plated.

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3.7 Identification marking. The optical inserts shall be legibly and permanently marked by metal stamping with the name or registered trademark of the manufacturer.

3.8 Workmanship. Optical inserts shall be from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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4.1.4 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. The unit of product for examination purposes shall be one insert.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective
Major A	II	1.0
Major B	S-3	1.0
Minor A	II	4.0
Dimensional	S-3	4.01/

1/ Except see Table III

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with Mil-STD-105, and as indicated in Table II. The unit of product for test purposes shall be one insert. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests.

Characteristics	Inspection level	AQL (percent defective
Metal composition	S-1	1.0
Plastic covering	S-1	1.0

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4.2.3 Component materials. Sufficient samples of the component materials shall be selected from each lot of materials used in the manufacture of the inserts to perform tests as required by this specification. When sample size for component materials is not specified, the sample size for each test shall be the quantity necessary to perform all required tests one time each. When possible, the same material may be used to perform two or more tests.

4.3 Examination. The inserts shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any inserts in the sample containing one or more defects shall be rejected and if the number of defective inserts in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.1 Classification of defects. Examination shall be conducted in accordance with but not restricted to the categories and defects in Table I. Additional defects shall be classified in accordance with MIL-STD-105.

Table III. Classification of defects.

Major A	
101	Component missing or of improper design.
102	Dimensions of binocular lens sub-assembly not in tolerance.
103	Eyewire not grooved properly.
104	When fitted with a standard lens template, eyewire clamp not completely closed.
Major B	
150	Struts improperly attached to eyewire.
151	Strut housing attached in wrong position.
152	Improper bridge attached.
153	Improperly assembled.
Minor A	
201	Metal parts not free of burrs or sharp edges.
202	Nickel plating missing.
203	Identification marking missing or in error.

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4.3.2 Dimensional examination. Any dimension not within the specified tolerance shall be classified as a defect, as specified in Tables I and III.

4.3.3 Lens fit. The lens described in 3.4.2 shall be inserted in the eyewire, and examination made when viewed in front of a bright light to assure that no spaces exist between the eyewire and lens.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis of metals. The samples, selected in accordance with 4.2.3, shall be analyzed for conformance with Method 111 or 112 of FED-STD-151, METAL: TEST METHOD. In case of dispute, Method 111 shall be used as the basis for acceptance.

4.5 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-D-129.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each optical insert shall be packaged in a plastic envelope of an appropriate size and design. Closure shall be as specified in the applicable box specification.

5.1.1.2 Intermediate package. Twelve unit packages containing optical inserts shall be packaged in a box conforming to PPP-B-566 or PPP-B-663, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for optical inserts shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packages, containing optical inserts shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

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5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Twenty-four intermediate packages containing optical inserts shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twenty-four intermediate packages containing optical inserts shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Twenty-four intermediate packages containing optical inserts shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceeds 250 pounds of 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 52 inches in width, and 43 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Optical Inserts packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Optical Inserts packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

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5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. This specification covers one type and class of optical inserts for use with Mask, Chemical-Biological, XM-40, by personnel requiring visual corrections.

6.2 Acquisition requirements.

a. Title, number, and date of the specification.

b. Type required (see 1.1).

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

d. National Stock Number.

6.3 National Stock Number (NSN) Coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6540-01-253-8169	INSERT, OPTICAL, CHEMICAL-BIOLOGICAL MASK, For use with XM-40 Mask.

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6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Optical Inserts
Chemical-Biological Mask
Frames
Binocular Lens

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6540-1207

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-42073	INSERT, OPTICAL, CHEMICAL-BIOLOGICAL MASK, XM-40

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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